

WASP-11b

R-0656

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_260205 · created 2026-07-27T17:22:52+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461076.7421 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.051
Transit depth (Rp/Rs)^2	2.99 +/- 1.77 [%]
Orbital Inclination (inc)	85.8 +/- 1.67
Airmass coefficient 1 (a1)	0.53 +/- 0.26
Airmass coefficient 2 (a2)	0.38 +/- 1.68
Scatter in the residuals of the lightcurve fit is	44.85 %
Best Comparison Star	None
Optimal Aperture	1.46
Optimal Annulus	6.25
Transit Duration (day)	0.046 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.051 vs 0.131 (deviation 0.82σ)
Duration fit vs published	0.046 d vs 0.1075 d (deviation 2.37σ)
Mid-time O–C	10.9 min
Depth SNR	3.4
Residual scatter	44.85 %

Light curve

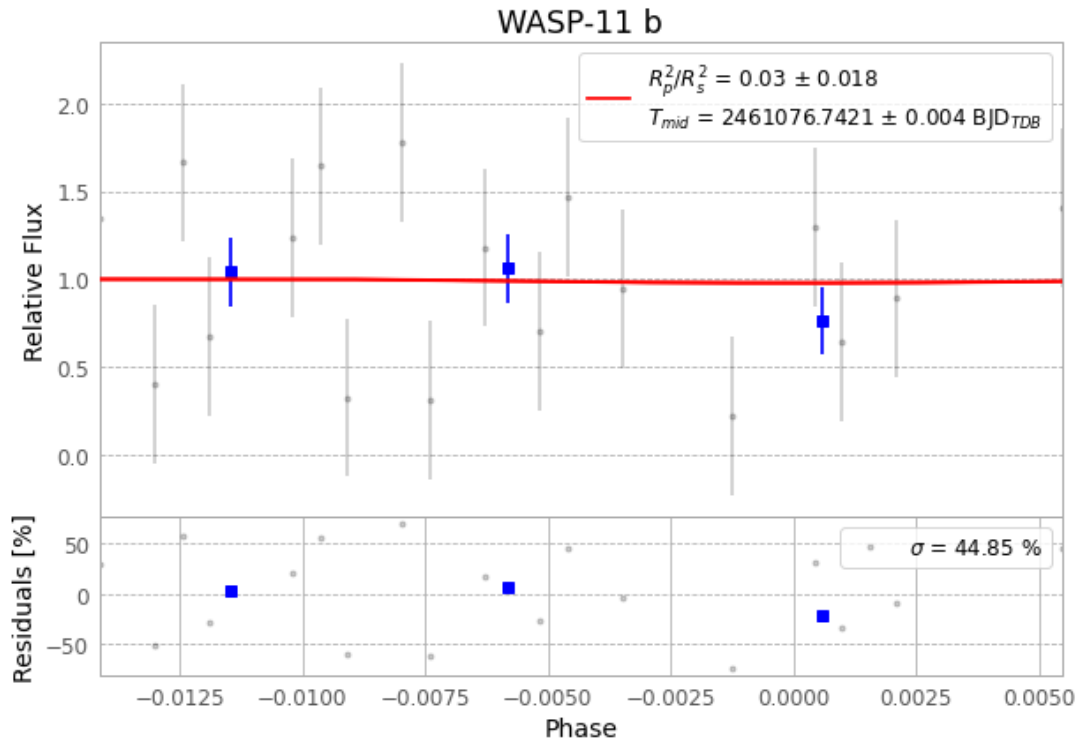


Figure 1 — FinalLightCurve_WASP-11 b_2026-02-04.png (SHA-256 fb10c86ed8ff781c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [396, 438], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5123c2ca45f299d7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 574e53e

Data provenance

72 FITS frames (47 MB), HATP-10260205042115.FITS → HATP-10260205075415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-260205.log (caba575d680c...), FinalLightCurve_WASP-11 b_2026-02-04.png (fb10c86ed8ff...), FinalLightCurve_WASP-11 b_2026-02-04.csv (ea29db1b27a2...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 17:22Z.